

Chapter – 10: Living Creatures: Exploring Their Characteristics

- Avadhi, Aayush – morning walk – their parents
- Avadhi – notice – shell – try – pick up – mother – advice – not to do – may be home – living snail
- Both – wonder – shell – not moving – living being inside
- Later in day – both – share incident – classmates – approach teacher
- Teacher – initiate discussion
- Activity –
 - Look around – obs. – list things – **identify** – living, non-living – your understanding
 - Also note – reason

What Sets the Living Apart from the Non-living

- May identify – movement – one characteristics – living
- BUT – cars – also move – are they living?
- Compare – yourself – car – what you can do BUT car cannot
- Yourself – excellent example – living being
- Grp. things – living or non-living – compare to yourself
- Car – does not grow – may be non-living
- Continue discussion – similar way – identify – essential characteristics – living beings
- Consider – **movement** – one characteristic – list 5 things – move independently
- These 5 things – are all living?
- BUT – plants – do not move – one place to another – are they living?
- Plants – do not move – BUT – flowers bloom – kind of movement
- Another example – insectivorous plants – depend on – insects – nutrition
- Drosera – ex. – insectivorous plants – saucer-shaped leaves – hair-like projections – unequal lengths – sticky ends
- Insect – enters saucer – hairs move inward – trap insects
- Climbers – wind themselves – around obj. – close to them
- Compare yourself – picture of childhood – you have grown
- Plants, living beings – grow – characteristics of living
- Living beings – food – **nutrition** – growth, development – list 5 living beings – require food
- Think of process – without which – cannot live
- Count – no. of breaths – per minute – after normal walk, after run, after dance
- Record and obs. – notice diff. – each situation
- Notice – breathing process – some animals – dogs, cats, cows, buffaloes – while rest
- Breathing – inhale – air – outside to inside body – exhale – air – inside to outside body
- Breathing – part of process – **respiration**
- Plants – also respire – tiny pores – stomata – surface of leaves
- These pores – help plants – breathe
- Notice – white patches – coloured shirts – under armpits – summer
- These patches – sweat – water and salt – waste products
- Removal – waste products – **excretion**
- Urine – also excreted – animals
- Plants – also excrete – excess water, minerals – small droplets – grasses, roses

- Another characteristic – **stimulus** – events – prompts living beings – **respond** immediately
- Ex. – stepping on thorn, touching hot obj.
- Plants – respond to stimuli – touch-me-not (*mimosa*, *chhui-mui*, *lajjalu*) – fold leaves – on touch
- Some plants – fold leaves – after sunset – *amla*
- All living beings – **respond to stimuli** – find some more – neighbourhood
- List young ones – 5 diff. animals
- All living beings – **reproduce** – process – produce – new one's – own kind
- Reproduction – necessary – continuity
- Living being – not exhibit – above characteristics – all resrc. – available – said to be **dead**
- Above discussion – conclude – **living** beings – common characteristics – show movements, need food, they grow, they respire, reproduce, excrete, respond to stimuli – eventually – die
- Absence – any characteristics – indicate – **non-living** beings
- Which category – seed – living OR non-living
- Let's explore

Essential Conditions for Germination of a Seed

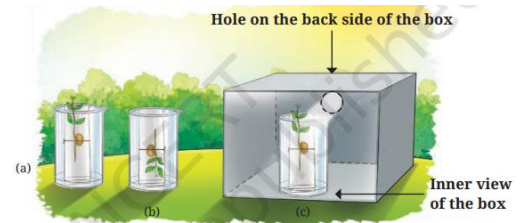
- Might wonder – what conditions – req. – **germination** – seed
- Activity –
 - Take – 4 identical pots – garden soil – sow 4 bean seeds – each pot – following conditions –
 - Pot A – no water – direct sunlight
 - Pot B – excess water – direct sunlight – keep adding water
 - Pot C – moderate water – complete dark – keep it moist
 - Pot D – moderate water – direct sunlight – keep it moist
 - Not down – availability – air, sunlight, water – each case
 - **Predict** – seed – germinate or not – each pot
 - Obs. – regularly – 7-10 days – check status – germination – record
 - Compare – records with predictions
- Compare – available conditions – each pot
- Germination – bean seeds – require – right amt. – water, air
- Water –
 - Seeds – req. – germination – carry out – necessary processes
 - Outer covering – seed – seed coat – water – softens it – helps tiny embryo – develop into plant
- Air and soil –
 - Seeds – need air – germination
 - Use air – spaces b/w soil – also allow – roots to grow
- Light / dark conditions –
 - Seeds – no light req. – germination
 - After germination – sunlight – req. – further growth
- Chapter – mindful eating – learnt – humans – req. – balanced diet
- Similarly – plants – need favourable cond., nutrients – proper growth, development

Growth and Movements in Plants

- Plants – respond to light?
- Dir. of growth – change? – inverted plants

- Activity –

- Take – bean, gram seeds – germinate – moist cloth, tissue paper
- Germinate until – develop into seedling – small root, shoot
- Take 3 tumblers – label – A, B, C
- Take – 3 glass plates – attach – thick blotting paper – each plate – thick soft cotton thread
- Fix – 1 seedling – each plate – ensure – plant – not damaged
- Place – 1 glass plate – with seedling – each beaker – A, C – as shown
- Beaker B – arrange plate – shoot – directed downwards – as shown
- Pour water – all 3 – ensure – seedling – above water
- Each case – bottom – blotting paper – get wet – seedling – gets water
- Beaker – A, B – sunlight – as shown
- Beaker C – cardboard box around – sunlight – small hole
- Note down – predictions and obs.



- Results – experiment above –

- Plant – upright – root – grow downward – shoot – grow upward
- Plant – inverted – root – grow downward – shoot – grow upward
- Plant – sunlight – one dir. – shoot – grow that dir. – root – still downward

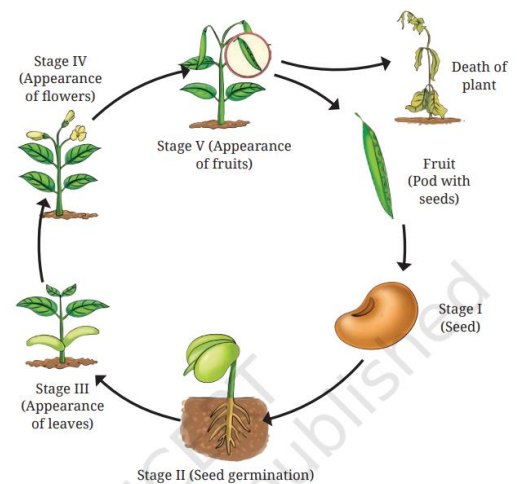
- Jagadish Chandra Bose – Indian scientist – fascinating exp. – plants

- Built machine – crescograph – record – plants' response – stimuli – light, heat, elec., gravity

Life Cycle of a Plant

- Activity –

- Plant – bean seed – proper cond. – growth – obs. – 3 months
- Record obs. – when changes – visible
- Record obs. – following ques. –
 - How long – any change occurs? – make sketches
 - First flower – how many days?
 - After flower dries – any further growth?
 - Which structure – remaining flower – develop to?
 - Notice – pod, fruit with seeds?
 - What happens to plant – after fruits are formed?



- Go through – obs. – what changes – after fruits?
- Sow seeds – watch – new gen.
- Seed grows – young plant – matures – produce flowers, fruits
- Fruit – pod – contain seeds – give rise – new gen.
- Entire process – seeds to new plant – new plant to new gen. – **life cycle**
- Plant – stops growing – all necessary cond. – still meet – considered dead

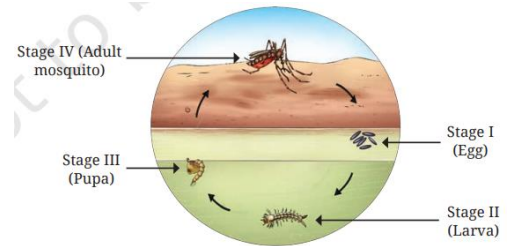
Life Cycle of Animals

- Learnt – life cycle – plants

- Lets obs. – animals' life cycle

Life cycle of a mosquito

- Mosquitoes around – common exp.
- Female mosquito – blood sucking insects – malaria, dengue, chikungunya
- Advised – not allow water – stagnate anywhere
- Conduct – safety audit – school, home, surrounding – check stagnant water
- Common places – stagnant water – desert coolers, planted pots, any other open containers
- 2 diff. worm like creatures – **larva**, **pupa** – diff. stages – life cycle – mosquito
- Discuss – teachers, other adults – measures – stop breeding
- Larvae, pupae – come to surface – need air – respire
- Activity –
 - Take – water container – mosquito larvae, pupae
 - Separate 4-5 larvae, pupae – 2 diff. containers
 - Obs. – everyday – until change
 - Larvae to pupae – larval stage 1st – else – vice-versa
- This act. – learn – sequence of growth
- Mosquitoes – 4 stages – life cycle – egg, larva, pupa, adult
- Adult mosquito – emerge from pupa – rest on surface – flies away
- Mosquito – begins life – egg – develops into – larva – grows into – pupa – transforms into – adult mosquito
- Significant – changes – appearance, body shape, structure – various stages



Life cycle of a frog

- Activity –
 - Awadhi, Aayush – dressed – full sleeves shirts, full pants – raining too much
 - They – going out – with classmates
 - Brief walk – science teacher – reach – shallow pond
 - Teacher – cautions them – watch from dist.
- May notice – white jelly like substance – surface of water
- Jelly-like structure – cluster of eggs – **spawn**
- Obs. features – all stages – shown in fig. – note down
- Some stages – clubbed together
- Life cycle – egg – progresses to – embryo – changes to **tadpole** – early stage – tail and no legs – later stage – hind legs – goes on to **froglet** – then – adult frog
- Obs. – tadpoles – develop legs – BUT – still contain tail – helps in swimming
- Tadpoles – grow gradually – look like frogs – froglets
- Froglets – continue to grow – legs – become strong

